Rohan Dhanraj Yadav

AI/ML ENGINEER · GENERATIVE AI · AGENTIC SYSTEMS · AI PLATFORM DEVELOPER

rohan.dhanraj.y@gma... | +91-7008958143 | Bengaluru, India

linkedin.com/in/rohan... | github.com/rohandha... | rohandhanraj.github.io

7+	4+	1.5y	3y	8+	15	6+
TOTAL EXPERIENCE	AI/ML EXP	AGENTIC AI	GEN AI	AI PROJECTS	TEAM SIZE	F500 CLIENTS

PROFESSIONAL PROFILE

Results-driven AI/ML Engineer with **7+ years of professional experience** including **4+ years specializing in designing, developing, and deploying production-grade AI systems** across enterprise domains. Deep expertise in **Generative AI, Agentic AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Multi-Agent Orchestration,** and **Machine Learning**.

Proven track record in leading cross-functional teams of **15+ engineers**, architecting scalable ML/AI pipelines, and delivering high-impact AI applications for **Fortune 500 clients** including Cisco, ITC Fairview, KDP, Omnitrax, PDS, and Marco Technologies. Known for building production systems handling **4,500+ daily API requests**, processing **100K+ daily records**, and achieving **18–38% model performance improvements** in production environments.

TECHNICAL SKILLS

LANGUAGES & FRAMEWORKS

Python, Django, Flask, FastAPI, Streamlit

GENERATIVE AI & LLM

LLMs, OpenAI GPT-4, LangChain, LangGraph, Prompt Engineering, RAG, Prompt Chaining, Chain-of-Thought Reasoning

AGENTIC AI SYSTEMS

Agent Orchestration, LangGraph Workflows, Multi-Agent Systems, Tool Integration, Memory Management, Autonomous Decision-Making

ML & DEEP LEARNING

Scikit-Learn, TensorFlow, PyTorch, Optuna, HuggingFace Transformers, Model Fine-tuning, Optimization, PyCaret

NLP SPECIALISTS

Spacy, NLTK, BERT, RoBERTa, CLIP, Flan-T5, Sentiment Analysis, NER, Text Classification

DATA ENGINEERING

Apache Airflow, ETL Processes, MySQL, PostgreSQL, MongoDB, Cassandra, Vector DBs (Qdrant, LanceDB, Chroma)

ANALYTICS & STATS

NumPy, Pandas, Matplotlib, Seaborn, Plotly, SciPy, Statistical Analysis, Predictive Analytics, Visualization

WEB SCRAPING

BeautifulSoup, Scrapy, Selenium, Playwright, Pydantic Structured Output

DEVOPS & CLOUD

Git, GitHub, Docker, CI/CD (GitHub Workflows, Jenkins), GCP, AWS, MLOps

WORK EXPERIENCE

AI Engineer — Backend & AI Systems Architecture

November 2024 – October 2025

AI Tech Solutions Ltd. | Full-time

Focus: Architecting production-ready Agentic AI platforms, advanced RAG systems, and multi-agent orchestration

- Led 15-member team building scalable AI systems using Python, FastAPI for backend services and production deployment
- Developed modular ML pipelines (Omodore platform) achieving **38% accuracy improvement** through advanced RAG architecture
- Designed multi-agent orchestration systems using **LangGraph** for complex workflow automation and autonomous decision-making
- Implemented prompt chaining and **chain-of-thought reasoning** mechanisms
- Deployed containerized AI systems using **Docker** with vector databases (Qdrant, LanceDB, Chroma)
- Built automated web scraping systems (Rbetraj) with **Playwright** and **Selenium** for real-time data extraction
- Architected multi-modal virtual agent (GALAMBO) integrating GPT-4 and autonomous tool use

`Python` `LangChain` `LangGraph` `FastAPI` `LLMs` `RAG` `Docker` `Vector DBs`

Software Engineer — Generative AI | ML | Data Science | Python

September 2022 – October 2024

Aimlytics Technology | Full-time

Focus: Enterprise-scale Generative AI, NLP, and ML solutions for Fortune 500 clients

- Led team of 8 engineers delivering AI solutions for **Fortune 500 clients**
- Built RAG-based virtual sales assistant (SalesMoji) handling **4,500+ daily API requests**
- Built ML pipelines processing **100K+ daily records** using **Apache Airflow**
- Fine-tuned **BERT** and **RoBERTa** models achieving **18% performance improvement**
- Implemented vector database solutions (Qdrant, Chroma) for semantic search
- Built multi-modal educational AI system (InventtEd)
- Developed multilingual NLP pipeline supporting **15+ languages** (evalmyBRAND)
- Resolved **28 critical production issues** with 100% on-time delivery

`Airflow` `BERT` `RoBERTa` `FastAPI` `GCP` `Qdrant`

Machine Learning Engineer Intern

August 2021 – September 2022

iNeuron.ai | Internship

Focus: ML pipeline development and optimization

- Built ML pipeline architectures for multi-class classification
- Reduced training time by **35%** using **Optuna**
- Deployed ML applications using Flask and Docker
- Built protein classification system (77 markers, 8 classes)

`Scikit-Learn` `Optuna` `Flask` `Docker` `AWS`

Metallurgical Quality Engineer

August 2016 – November 2019

Bureau Veritas Group | Full-time

Focus: Quality assurance and industrial data analytics

- Conduct metallurgical testing and quality assurance across major steel companies
- Apply Python (NumPy, Pandas, SciPy, Plotly) for statistical analysis and data-driven quality improvement
- Manage quality compliance and predictive analysis for multiple production facilities

PROJECT SYNOPSIS

Project 1: OMODORE

Nov 2024 - Oct 2025

Domain: AI Agent Platform | Role: Senior AI Engineer

Omodore is an auto-agent platform enabling non-technical users to build domain-specific AI agents without coding expertise. The platform leverages advanced RAG pipelines to deliver accurate, context-aware responses with a 45% accuracy boost.

- Led 15-member team developing Gen AI and Agentic AI solutions
- Architected scalable RAG pipelines using LangChain and LangGraph
- Implemented prompt chaining and Chain of Thoughts methodology for improved reasoning
- Built automated agent building workflows using FastAPI
- Integrated multiple LLMs for optimal performance and Qdrant/Milvus for retrieval

`Python` `FastAPI` `LangGraph` `RAG` `Docker`

Project 2: RBETRAJ

Nov 2024 - Oct 2025

Domain: E-commerce Data Intelligence | Role: Senior AI Engineer

Rbetraj is an Automated Dropshipping Platform powered by Shop Scan Agent specializing in e-commerce data extraction with advanced price tracking. Includes open-source scraper AI TechCrawler for extraction and insights.

- Developed open-source automated scraping framework AI Tech Crawler
- Implemented GPT-powered and LLM-based structured data extraction with Pydantic

- Created robust scraping pipelines using Playwright and BeautifulSoup
- Designed RESTful APIs using FastAPI for distributed data access

`Playwright` `FastAPI` `Pydantic` `BeautifulSoup` `Docker`

Project 3: GALAMBO

Nov 2024 - Oct 2025

Domain: Multi-Modal AI Assistant | Role: Senior AI Engineer

Sophisticated multi-modal virtual agent combining context-aware GPT responses with advanced image search and autonomous tool use.

- Architected multi-modal AI system integrating text and image processing
- Implemented autonomous tool use and search engine integration for information retrieval
- Built scalable API infrastructure using FastAPI and optimized model response times

`Multi-modal AI` `GPT-4` `LangChain` `FastAPI`

Project 4: SALESMOJI

Nov 2023 - Oct 2024

Domain: Sales and Marketing | Role: Software Engineer (Gen AI | ML)

Harnesses GPT and LLMs to answer sales questions in plain English. Processes 1M+ daily records through Airflow pipelines.

- Led 8-member team delivering custom AI workflows for Fortune 500 clients
- Built RAG-based Virtual Sales Assistant handled 10K+ daily queries with 89% accuracy
- Architected ML pipelines for predictive sales forecasting and trend analysis
- Implemented vector DB solutions (Qdrant, Milvus, Weaviate) for efficient retrieval

`GPT` `RAG` `Airflow` `Milvus` `TensorFlow`

Project 5: AISERA

July 2023 - Oct 2024

Domain: Customer Service and IT Operations | Role: Software Engineer (AI | ML)

Enterprise Generative AI platform for automating customer support through NLP and ML workflows.

- Led team of 8 (5 Data Eng + 3 Soft Dev) delivering workflows for Cisco, ITC, etc.
- Fine-tuned BERT models achieving 50% performance improvement in sentiment analysis
- Implemented custom field prediction using advanced BERT architectures and Airflow triggers
- Architected scalable ML pipelines and integrated LangChain for conversational AI

`BERT` `LangChain` `Chroma` `FastAPI` `GCP`

Project 6: EVALMYBRAND

Sep 2022 - Oct 2024

Domain: Customer Experience | Role: Software Developer (AI | ML)

AI-powered platform supporting 15+ languages to turn unstructured feedback into brand insights and sentiment tracking.

- Fine-tuned RoBERTa for Aspect-Based Sentiment Analysis with 50% performance gain
- Implemented BERT for Neural Machine Translation with 27% accuracy improvement
- Built innovative pipeline for mixed-code translation using Google/LibreOffice APIs
- Orchestrated large-scale data pipelines using Apache Airflow and Oracle/Cassandra

`RoBERTa` `NMT` `Airflow` `FastAPI` `Poetry`

Project 7: INVENTTED

Jan 2023 - Aug 2023

Domain: EdTech | Role: Software Developer (AI | ML)

Pioneering EdTech platform providing AI-powered tools for summarization, essay grading, and performance prediction.

- Fine-tuned CLIP for text-to-image generation and Flan-T5 for keyword-based MCQ generation
- Designed scaled Phishing Classifier and multi-modal social media classification model
- Integrated multiple AI models into a cohesive platform with optimized Transformers
- Performed advanced prompt engineering for ChatGPT-based virtual tutoring

`CLIP` `Flan-T5` `Prompt Eng` `GCP`

Project 8: MICE PROTEIN EXPRESSION

Aug 2021 - Sep 2022

Domain: Bioinformatics | Role: ML Engineer Intern

Multi-class classification of 77 protein markers in Down syndrome research involving complex biological datasets.

- Led team of 10 to architect end-to-end ML system with automated pipelines
- Achieved 40% training time reduction using Optuna hyperparameter tuning
- Developed large-scale biological data preprocessing and Flask-based model serving
- Integrated MySQL and Cassandra DB for efficient data management

`Optuna` `Flask` `Cassandra` `PyCaret` `AWS`

EDUCATION

B.Tech. Mechanical Engineering

2016

Biju Patnaik University of Technology, Odisha

GATE Qualified — Credential ID: A2WR3

CERTIFICATIONS

🏆 **GATE ME 2016**
Credential ID: A2WR3

🏆 **Machine Learning Zoom Camp**
DataTalksClub | Credential ID: 8F163B

🏆 **Gold Badge: Python**
HackerRank | Credential ID: 6920934A49FA

🏆 **Gold Badge: Problem Solving**
HackerRank | Credential ID: 113775E95540

🏆 **10 Days of Statistics**
Gold Badge | HackerRank

🏆 **30 Days of ML Challenge**
Kaggle | Rank 69 / 7,573 teams

🌟 ACHIEVEMENTS & AWARDS

⚡ **7+ years** of professional experience including 4+ years in AI/ML engineering

⚡ Achieved **Gold Badges** for Python, Problem Solving, and Statistics on HackerRank

⚡ Secured **Rank 69 out of 7,573 teams** in Kaggle ML Challenge

⚡ Led teams of **8–15 members** across enterprise AI projects

⚡ Delivered **50% model performance improvements** in production systems (BERT, RoBERTa)

- ⚡ Built systems handling **4,500+ daily API requests**
- ⚡ Processed **100K+ daily records** via Airflow pipelines
- ⚡ Achieved **38% accuracy boost** in RAG systems
- ⚡ Reduced training time by **35%** using Optuna optimization
- ⚡ Resolved **28 critical production issues** across Fortune 500 client engagements